

Department of Physical Science
Faculty of Applied Science
IT4216 Research Project
Undergraduate Research Synopsis Submission Form

Student Information

Name	R.M.S.Rathnayake
Reg. No	2020/ICT/47
Degree Program	B.Sc. (Hons) in Information Technology
Email	maleesharathnayake840@gmail.com
Phone No	+94704169125

Research Details

Proposed Research Title	ConstRSA: A Minimal Constant-Time RSA-512 Digital Signature with Statistical Timing Leak Verification on Standard Laptops
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Brief Description of the Research (100-150 words):

RSA digital signatures are widely used for secure communications, but many software implementations suffer from timing leaks that can expose private keys. This research aims to develop and verify a minimal, constant-time RSA-512 digital signature algorithm named ConstRSA. The implementation will use constant-time techniques such as Montgomery multiplication and bit-masking to eliminate secret-dependent timing variations. Statistical verification will be carried out using Welch's t-test on large datasets to ensure no timing correlation with secret keys. The study will benchmark ConstRSA's performance and release the verified implementation as open-source for educational and prototyping use.

Objectives of the Research:**Primary Objective**

To design, implement, and statistically verify a minimal constant-time RSA-512 digital signature (PSS + SHA256) in under 10 KB, executable and testable on standard consumer laptops.

Specific Objectives

1. Implementation: Develop a branchless RSA-512 signature in C using CRT, Montgomery multiplication, and fixed-window exponentiation.
2. Constant-Time Enforcement: Eliminate all secret-dependent branches and memory access patterns using bit-masking and fixed loops.
3. Statistical Verification: Collect timing traces and apply Welch's t-test to prove no correlation with secret key bits ($p > 0.05$).
4. Performance Benchmarking: Measure signing/verification time and binary size on the same hardware.
5. Open-Source Release: Publish full code, tests, and documentation on GitHub for academic and practical reuse.

Student Declaration:

I confirm that the contents of this proposal synopsis have been developed independently and have not been copied, in whole or in part, from any other source.

Student's Signature

Date

Supervisors Details

Primary Supervisor Name

Mr. B. Yogarajah

Co-supervisors
(Name & Institution/Address)

1. Mr. K. Mathanaharan

2. Dr.S.Kirushanth

Primary Supervisors Approval

Requires Revision

Comments (if any):

Primary Supervisor Signature

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Approval of Head

Approved

Rejected

Requires Revision

Comments:

Signature of Head of the Department

Date